# Supplementary Material for
# Modality-Aware Shot Relating and Comparing for Video Scene Detection

The following results are obtained in a fully supervised manner on the MovieNet (Huang et al. 2020) dataset, unless otherwise specified.

## 1 Additional Experimental Results

### 1.1 Ablation Studies on Place Continuity Graph

**Detail-Wide Shot Affiliation.** The composition of detail-wide shot affiliation, as defined in Eq. (4), is ablated in Table S1. The results indicate that combining shot similarity and temporal proximity yields optimal performance. This underscores the importance of considering both shot similarity and temporal proximity when determining the affiliation between wide shots and detail shots.

| Local-Global Shot Affiliation | AP | mIoU | F1 |
|---|---|---|---|
| Shot Similarity $S^{\mathrm{P}}_{ij}$ | 65.2 | 64.7 | 63.3 |
| Temporal Proximity $D_{ij}$ | 63.9 | 60.1 | 59.9 |
| Both | 67.4 | 65.8 | 63.8 |

Table S1: Comparison of variants of local-global shots affiliation as defined in Eq. (4).

**Necessary of Using Both Detail-to-Wide (D2W) and Wide-to-Detail (W2D) Graphs.** Table S2 presents the results of assessing the contributions of different components of our proposed PCG to overall performance. Combining D2W and W2D achieves significantly better performance compared to using only D2W or only W2D.

| D2W | W2D | AP |
|---|---|---|
| ✓ | - | 63.6 |
| - | ✓ | 65.3 |
| ✓ | ✓ | 67.4 |

Table S2: Ablation study on PCG graph in different inclusions of D2W and W2D graphs. ✓signifies "included", while - "excluded".

**Visualization of Edge Connections in EJG and PCG.** In Fig. S1, we present a sample result of edge connections of EJG and PCG, built upon Sec. 3.2. As expected, EJG connects similar but distant shots, while PCG links consecutive shots depicting the same location. Consequently, EJG is advantageous for modeling long-range shot relations, whereas PCG is effective for modeling short-range shot relations.

### 1.2 Comparison with MHRT under Various Modality Combinations

Fig. S2 compares the performance of our MASRC with MHRT[1] (Wei et al. 2023), a state-of-the-art multi-modal video scene detection method, under the same entity and place feature inputs. Our model beats MHRT regardless of the combination of different visual semantic modalities. This superiority of ours is mainly thanks to diverse shot relationship modeling and context comparison convolution for ending shot detection.

[1]The code for MHRT is not available; hence, we reproduced it for comparisons.

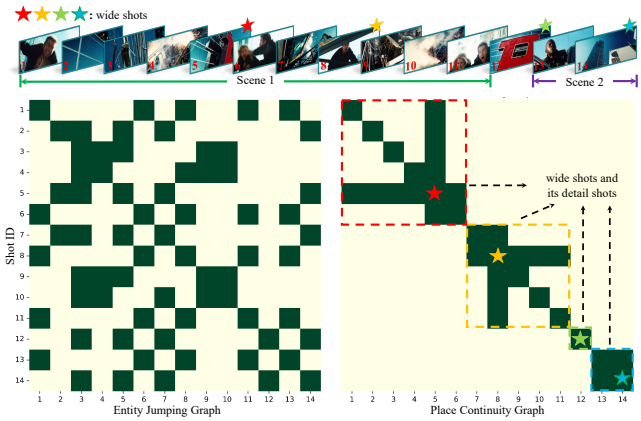


Figure S1: Visualization of edges connections (in dark color) of EJG and PCG. Stars indicate different wide shots discovered by the place modality, and the others are detail shots.

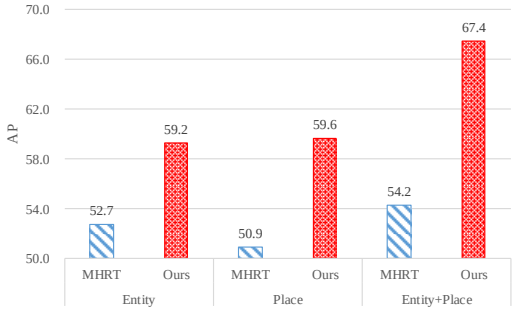


Figure S2: Comparison between proposed MASRC and MHRT (Wei et al. 2023) under various combinations of multi-modality semantic inputs.

### 1.3 Computation Cost Analysis

Under identical hardware conditions, Table S3 presents the number of training parameters, the training/inference throughput measured in samples per second (Sam./s), and GPU memory costs per sample (GB/Sam.) for state-of-the-art methods. A sample corresponds to a time window with a length of 14 shots, equating to 14 nodes in our graph. As can be seen, our model requires fewer resources in terms of training params, training/inference throughput, and GPU memory costs compared to others utilizing the same backbone.

Figure S3: Qualitative comparison of the proposed method with the previous method MHRT (Wei et al. 2023). GT denotes ground-truth scene boundaries for reference. The colored vertical lines represent the video scene boundaries.

Figure S4: Qualitative comparison of the proposed method with the previous method MHRT (Wei et al. 2023). GT denotes ground-truth scene boundaries for reference. The red vertical lines represent the video scene boundaries.

| Methods | Modalities | Train Par. | Train | | Inference | |
|---|---|---|---|---|---|---|
| | | | Sam./s | GB/sam. | Sam./s | GB/sam. |
| LGSS (Rao et al. 2020) | P | 13.7 M | 32 | 0.37 | 68 | 0.17 |
| Ours | P | 12.9 M | 3788 | 0.0013 | 5632 | 0.0005 |
| MHRT (Wei et al. 2023) | E+P | 27.8 M | 2509 | 0.0020 | 3584 | 0.0014 |
| Ours | E+P | 21.4 M | 3093 | 0.0019 | 5171 | 0.0009 |

Table S3: Number of training parameters (Train Par.), training/inference throughput measured in samples per second (Sam./s), and GPU memory costs per sample (GB/Sam.) for state-of-the-art methods. E is short for entity modality and P is for place modality.

## 1.4 Visualization of Short Scene Detection.

In Fig. S3, we present a sample results of video scene detection on the MovieNet dataset. Ground-truth annotations are provided for reference, and we include the previous state-of-the-art method, MHRT (Wei et al. 2023), for comparison. The results demonstrate that MHRT is susceptible to camera shot changes, including switching, zooming, and movement, leading to over-segmented video scenes. In contrast, our proposed method effectively handles these challenging cases without experiencing over-segmentation. This is attributed to our approach on establishing the diverse relations between shots and carefully comparing the relations between shots.

## 1.5 Visualization of Long Scene Detection.

Fig. S4 depicts a result of long scene consisting of 76 shots. It portrays a taxi driver picking up two separate groups of passengers during his night shift before finishing his work. The sequence involves numerous rapid shot transitions. Previous SOTA (Wei et al. 2023) incorrectly segments the scene into 12 scenes, while our method reduces false positive detection and segments the scene into 5 scenes. This is attributed to our approach on establishing modality-aware shot relations and comparing the relations between shots.

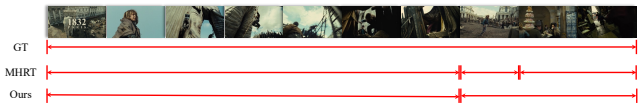


Figure S5: Qualitative comparison of the proposed method with the previous method MHRT (Wei et al. 2023). GT denotes ground-truth scene boundaries for reference. The red vertical lines represent the video scene boundaries.

## 1.6 Limitation.

We find some commonalities in cases where our method and previous SOTA method (Wei et al. 2023) may fail. Fig. S5 presents a short scene from "Les Misérables". Both our method and previous SOTA (Wei et al. 2023) segment this scene into two parts. The inconsistency between shots arises from frequent motion transitions, which weaken the coherence of entities and places, leading to incorrect segmentation. Addressing this challenge may involve incorporating background audio, such as music, which can provide another type of contextual consistency. Exploring the integration of audio into our model is a future research direction.

## 1.7 Impact of MCD on Detecting Video Scenes of Different Lengths

Fig. S6 provides a deeper analysis of the impact of the proposed MCD on video scene detection. We let length 12 to distinguish between short scenes (each of which has $\leq 12$ shots) and long scenes (each of which has $> 12$ shots) based on the fact that the average shot number in MovieNet scenes is 12.6. As shown in Fig. S6, compared with the model without MCD, the model with MCD improves both short and long scene detection. Notably, in long scene detection, the inclusion of MCD (72.6% AP) significantly enhances the

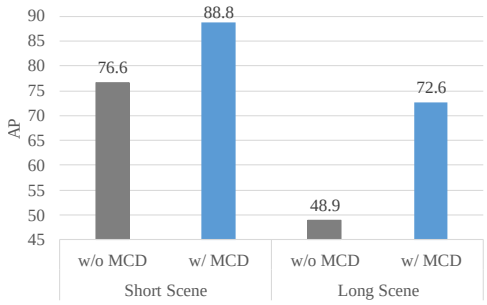


Figure S6: Influence of w/o and w/ MCD on detecting video scenes with varied lengths.

performance compared to the exclusion of MCD.

## References

Huang, Q.; Xiong, Y.; Rao, A.; Wang, J.; and Lin, D. 2020. MovieNet: A Holistic Dataset for Movie Understanding. In *ECCV*, volume 12349, 709–727.

Rao, A.; Xu, L.; Xiong, Y.; Xu, G.; Huang, Q.; Zhou, B.; and Lin, D. 2020. A Local-to-Global Approach to Multi-Modal Movie Scene Segmentation. In *CVPR*, 10143–10152.

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